Algebra 2 — Unit 1: Linear Functions

Practice Test — Answer Key

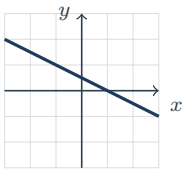
Instructions: Show all work. Write every line in $y = mx + b$ form when asked. No notes or calculator unless your teacher says so. Write each final answer in its box. Only what is inside a box is graded. **Total: 100 pts**

Keep in Mind

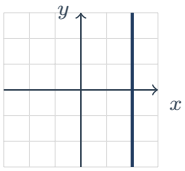
This is a practice test. It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers differ. Work every problem with no notes, then check your answers against the key.

Part A — Types of Slope (8 pts)

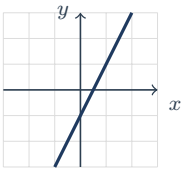
For each line, write whether its slope is **positive**, **negative**, **zero**, or **undefined**.



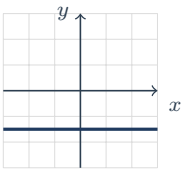
1. negative



2. undefined



3. positive



4. zero

No graph this time — decide from the points or the equation.

5. The line through $(3, 2)$ and $(-1, 2)$: **zero**

6. The line $y = 5x - 1$: **positive**

7. The line $x = -4$: **undefined**

8. The line through $(0, 6)$ and $(2, 1)$: **negative**

Part B — Writing Equations of Lines (20 pts)

Write each line in slope-intercept form, $y = mx + b$. Show how you found m and b; for 11 and 12, start in point-slope form, $y - y_1 = m(x - x_1)$.

9. The line through $(1, 5)$ and $(3, 9)$.

$$m = \frac{9 - 5}{3 - 1} = \frac{4}{2} = 2$$

$$5 = 2(1) + b \quad \Rightarrow \quad b = 3$$

Equation: $y = 2x + 3$

10. The line through $(-4, 5)$ and $(2, -4)$.

$$m = \frac{-4-5}{2-(-4)} = \frac{-9}{6} = -\frac{3}{2}$$

$$5 = -\frac{3}{2}(-4) + b = 6 + b \quad \Rightarrow \quad b = -1$$

Equation: $y = -\frac{3}{2}x - 1$

11. The line with slope 2 that passes through $(3, 4)$.

$$y - 4 = 2(x - 3)$$
$$y - 4 = 2x - 6$$
$$y = 2x - 2$$

Equation: $y = 2x - 2$

12. The line with slope $-\frac{1}{3}$ that passes through $(6, 1)$.

$$y - 1 = -\frac{1}{3}(x - 6)$$
$$y - 1 = -\frac{1}{3}x + 2$$
$$y = -\frac{1}{3}x + 3$$

Equation: $y = -\frac{1}{3}x + 3$

Part C — Piecewise Functions (20 pts)

13. **Evaluate.** (8 pts) $f(x) = \begin{cases} 3x + 2, & x < -1 \\ -1, & -1 \le x \le 1 \\ 2x - 5, & x > 1 \end{cases}$

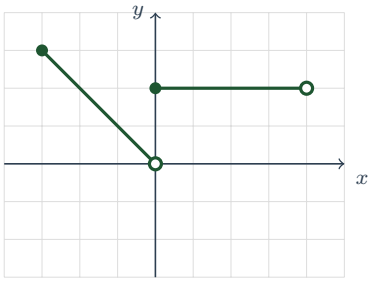
$f(-2) =$ -4 $\quad f(-1) =$ -1 $\quad f(1) =$ -1 $\quad f(4) =$ 3

14. **Domain.** (12 pts) Give the domain of each function as an inequality or in interval notation.

(a) $g(x) = \begin{cases} x - 2, & x < 3 \\ 7, & 3 \le x < 6 \end{cases}$ Domain: $x < 6$, $(-\infty, 6)$

(b) $h(x) = \begin{cases} x + 1, & -2 \le x \le 0 \\ 4 - x, & 0 < x \le 5 \end{cases}$ Domain: $-2 \le x \le 5$, $[-2, 5]$

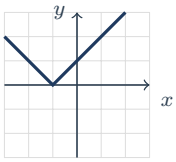
(c) The function k graphed below.



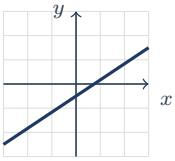
Domain: $-3 \le x < 4$, $[-3, 4)$

Part D — The Absolute Value Function (28 pts)

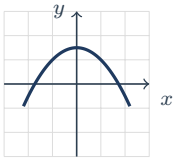
15. (3 pts) Write the letter of the graph of an **absolute value** function in its Letter box. **Letter:** **A**



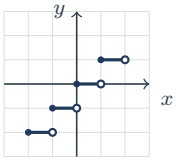
→ **(A)**



(B)



(C)



(D)

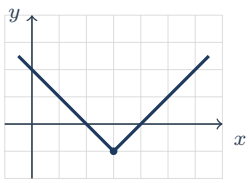
16. (3 pts) Which equation is an **absolute value** function? **Letter:** **C**

(A) $y = x^2 - 3$ (B) $y = 2x + 5$ → **(C)** $y = |x + 1| - 3$ (D) $y = \lfloor x \rfloor + 1$

17. (15 pts) Each function is a transformation of the parent $y = |x|$. Fill in the table.

Function	Vertex	Left/right shift	Up/down shift	Opens
$g(x) = \lvert x + 2 \rvert + 3$	$(-2, 3)$	**left 2**	**up 3**	**up**
$h(x) = \lvert x - 5 \rvert - 4$	$(5, -4)$	**right 5**	**down 4**	**up**
$k(x) = -\lvert x \rvert - 2$	$(0, -2)$	**none**	**down 2**	**down**

18. (4 pts) Which equation matches the graph? **Letter:** **C**



(A) $y = |x + 3| - 1$
(B) $y = |x - 3| + 1$
→ **(C)** $y = |x - 3| - 1$
(D) $y = -|x - 3| - 1$

19. (3 pts) Compared with the parent $y = |x|$, the graph of $y = \frac{1}{2}|x|$ is **Letter:** **B**

(A) narrower → **(B) wider** (C) shifted down 2 (D) shifted left 2

Part E — Absolute Value Equations & Inequalities (24 pts)

Solve. Show both cases.

20. (8 pts) $|2x + 1| = 9$

$$\mathbf{2x + 1 = 9 \quad or \quad 2x + 1 = -9}$$
$$\mathbf{2x = 8 \quad or \quad 2x = -10}$$

Solution: $x = 4$ **or** $x = -5$

21. (8 pts) $|x - 3| \leq 2$

$$\mathbf{-2 \leq x - 3 \leq 2}$$
$$\mathbf{1 \leq x \leq 5}$$

Solution: $1 \leq x \leq 5$

22. (8 pts) $|x + 4| > 6$

$$\mathbf{x + 4 > 6 \quad or \quad x + 4 < -6}$$
$$\mathbf{x > 2 \quad or \quad x < -10}$$

Solution: $x > 2$ **or** $x < -10$